

Mazur's Program B, after Furio–Lombardo

David Zureick-Brown (Amherst College)
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ChaBONNty, at the Max Plank Institute for Mathematics

July 2, 2026

Slides available at <https://dmzb.github.io/>

Summary of what's new:

- We solve the final two 7-adic cases of program B (in progress)
- using Furio–Lombardo's recent work, and
- a “variant” of Kummer's proof of FLT for regular primes

Condensed Pseudo-Perfectoid Higher ∞ -Stacks and a Derived Hyper-Monoidal Liquid Arithmetic Bhargava–Gauss–Langlands–Scholze–Rota–Witt Correspondence via δ -modularity of ℓ -adic forms on the Rigid Interuniversal Log-Crystalline Site of Non-Teichmüller Motives: A Gentle Introduction¹

David Zureick-Brown (Amherst College)

Santiago Arango-Piñeros (UMass Amherst)

$$\begin{array}{c}
 \begin{array}{ccc}
 \tilde{X}_A & \xleftrightarrow{\alpha_u} & \tilde{X}_A \\
 \downarrow \psi_B & \searrow f & \downarrow \psi_B \\
 \Pi_1(G) & \xrightarrow{\quad} & X_B
 \end{array}
 \end{array}
 \quad
 \begin{array}{c}
 D^+(R_{\infty}(1/p) \otimes \mathcal{L} \mathcal{C}^{\bullet}(S) = R\Gamma(\widehat{Y}_B, \widehat{\mathcal{D}}_{\text{LG}} \widehat{T}) \\
 - |x|_{\text{dd}} < e. \implies R\Gamma(\widehat{Y}'_B, \widehat{\mathcal{D}}_{\text{LT}}) \approx \text{RHom}(S, T) \\
 \sum_z i\{n \text{ Ext}_1^z(M, N) = \lfloor \frac{\mathcal{T}_c}{\mathcal{J}} \cdot \text{Gal}(L/M) \triangleleft \text{GL}_2(\dot{\mathbb{E}})^{L+}
 \end{array}
 \quad
 \begin{array}{c}
 \text{Graph of } y = \sin(x) \text{ with area under curve shaded} \\
 \tilde{\Pi}_{\text{GB}} = \frac{M}{\mathcal{T}_E n} \\
 \lim_{\alpha \rightarrow \infty} = E_k^r \quad \tau_n = \text{Tr}_{\mathcal{O}}(\Phi^{\mathcal{C}_1} V_1) \rightarrow 0 \rightarrow F_E \rightarrow C^l \rightarrow 0 \\
 \frac{\widehat{\mathcal{I}}_c}{\mathcal{I}_{\widehat{\mathcal{I}}+H}} = \frac{+H}{\mathcal{I}_{\widehat{\mathcal{I}}+H}} \\
 \text{Graph of } y = \sin(x) \text{ with area under curve shaded}
 \end{array}
 \quad
 \begin{array}{c}
 |D_j^a| > \text{RH}^{\Delta}(\widehat{V}_{\psi}, \overline{V}) - H_{\frac{u}{\theta}}^u(X, \overline{Z}_p) = (\widehat{T}_p)^{\frac{n}{2}} \parallel \text{Mod}_{\Delta} \Leftrightarrow \widehat{D}F\text{-Isoc } \acute{o}.
 \end{array}$$

¹but with Full Proofs

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Galois Representations

$$\begin{aligned}\mathbb{Q} &\subset K \subset \overline{\mathbb{Q}} \\ G_K &:= \text{Aut}(\overline{K}/K) \\ E[n](\overline{K}) &\cong (\mathbb{Z}/n\mathbb{Z})^2\end{aligned}$$

$$\begin{aligned}\rho_{E,n}: G_K &\rightarrow \text{Aut } E[n] \cong \text{GL}_2(\mathbb{Z}/n\mathbb{Z}) \\ \rho_{E,\ell^\infty}: G_K &\rightarrow \text{GL}_2(\mathbb{Z}_\ell) = \varprojlim_n \text{GL}_2(\mathbb{Z}/\ell^n\mathbb{Z}) \\ \rho_E: G_K &\rightarrow \text{GL}_2(\widehat{\mathbb{Z}}) = \varprojlim_n \text{GL}_2(\mathbb{Z}/n\mathbb{Z})\end{aligned}$$

Image of Galois

$$\rho_{E,n}: G_{\mathbb{Q}} \twoheadrightarrow H(n) \hookrightarrow \mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z})$$

$$\left\{ \begin{array}{c} \overline{\mathbb{Q}} \\ | \\ \overline{\mathbb{Q}}^{\ker \rho_{E,n}} = \mathbb{Q}(E[n]) \\ | \\ \mathbb{Q} \end{array} \right\} H(n)$$

Problem (Mazur's "program B")

Classify all possibilities for $H(n)$.

Mazur's Program B

As presented at Modular functions in one variable V in Bonn

Theorem 1 also fits into a general program:

B. Given a number field K and a subgroup H of $GL_2(\hat{\mathbb{Z}}) = \prod_p GL_2(\mathbb{Z}_p)$ classify
all elliptic curves E/K whose associated Galois representation on torsion points
maps $\text{Gal}(\bar{K}/K)$ into $H \subset GL_2(\hat{\mathbb{Z}})$.

Mazur - Rational points on modular curves (1977)

Example - torsion on an elliptic curve

If E has a K -rational **torsion point** $P \in E(K)[n]$ (of exact order n) then:

$$H(n) \subset \begin{pmatrix} 1 & * \\ 0 & * \end{pmatrix}$$

since for $\sigma \in G_K$ and $Q \in E(\overline{K})[n]$ such that $E(\overline{K})[n] \cong \langle P, Q \rangle$,

$$\begin{aligned}\sigma(P) &= P \\ \sigma(Q) &= a_\sigma P + b_\sigma Q\end{aligned}$$

Example - Isogenies

If E has a K -rational, **cyclic isogeny** $\phi: E \rightarrow E'$ with $\ker \phi = \langle P \rangle$ then:

$$H(n) \subset \begin{pmatrix} * & * \\ 0 & * \end{pmatrix}$$

since for $\sigma \in G_K$ and $Q \in E(\overline{K})[n]$ such that $E(\overline{K})[n] \cong \langle P, Q \rangle$,

$$\begin{aligned} \sigma(P) &= a_\sigma P \\ \sigma(Q) &= b_\sigma P + c_\sigma Q \end{aligned}$$

Example - other maximal subgroups

$\mathbb{F}_{p^2}^*$ acts on $\mathbb{F}_{p^2} \cong \mathbb{F}_p \times \mathbb{F}_p$

Normalizer of a non-split Cartan:

$$C_{\text{ns}} = \text{im} \left(\mathbb{F}_{p^2}^* \rightarrow \text{GL}_2(\mathbb{F}_p) \right) \subset N_{\text{ns}}$$

$H(n) \subset N_{\text{ns}}$ and $H(n) \not\subset C_{\text{ns}}$ iff

E admits a “necklace” (Rebolledo, Wuthrich)

Modular curves

Definition

- $X(N)(K) := \{(E/K, P, Q) : E[N] = \langle P, Q \rangle\} \cup \{\text{cusps}\}$
- $X(N)(K) \ni (E/K, P, Q) \Leftrightarrow \rho_{E,N}(G_K) = \{I\}$

Let $\Gamma(N) \subset H \subset \text{GL}_2(\widehat{\mathbb{Z}})$. The minimal such N is the **level** of H .

Definition

$X_H := X(N)/H(N)$ (where $H(N)$ is the image of H in $\text{GL}_2(\mathbb{Z}/N\mathbb{Z})$)

$$X_H(K) \ni (E/K, \iota) \Leftrightarrow \rho_{E,N}(G_K) \subset H(N)$$

Stacky disclaimer

This is only true up to twist; there are some subtleties if

- 1 $j(E) \in \{0, 12^3\}$ (plus some minor group theoretic conditions), or
- 2 if $-I \in H$.

Rational Points on modular curves

Mazur's program B

Compute $X_H(\mathbb{Q})$ for all H .

Remark

- Sometimes $X_H \cong \mathbb{P}^1$ or elliptic with rank $X_H(\mathbb{Q}) > 0$
- Some X_H have **exceptional** points
(i.e. $X_H(\mathbb{Q})$ is finite and E is non-cuspidal and non-CM)
- Can compute $g(X_H)$ group theoretically (via Riemann–Hurwitz).

Fact

$$g(X_H), \gamma(X_H) \rightarrow \infty \text{ as } [\mathrm{GL}_2(\widehat{\mathbb{Z}}) : H] \rightarrow \infty.$$

Let E be an elliptic curve over $\mathbb{Q}$ without CM

Uniformity conjecture/Question of Serre

For $\ell > 37$, $\rho_{E,\ell}$ is surjective.

Conjecture (Zywina)

$$\left[\mathrm{GL}_2(\widehat{\mathbb{Z}}) : \rho_E(G_{\mathbb{Q}}) \right] \leq 2736.$$

Zywina: the indices occurring infinitely often divide

220, 336, 360, 504, 864, 1152, 1200, 1296 or 1536

(modulo conjectures)

Remark

$j(E) \in \{-7 \cdot 11^3, -7 \cdot 137^3 \cdot 2083^3\}$ *does occur finitely often*

Recall

A point $(E, \iota) \in X_H(K)$ is **exceptional** if $X_H(K)$ is finite and $\text{End } E = \mathbb{Z}$.

Let ℓ prime, $E/\mathbb{Q}$ be a non-CM elliptic curve, and $H = \rho_{E, \ell^\infty}(G_{\mathbb{Q}})$.

Theorem (Rouse–Sutherland–ZB 2021)

Exactly one of the following is true:

- ① $X_H(\mathbb{Q})$ is infinite and H is listed in (Sutherland–Zywina 2017);
- ② X_H has a rational exceptional point listed in Table 1;
- ③ $H \leq N_{\text{ns}}(3^3), N_{\text{ns}}(5^2), N_{\text{ns}}(7^2), N_{\text{ns}}(11^2)$, or $N_{\text{ns}}(\ell)$ for some $\ell > 13$; or
- ④ $H \leq G_{\text{sp}}^\#(7^2), G_{\text{ns}}^\#(7^2)$ (49.179.9.1 or 49.196.9.1).

Conjecture

Cases (3) and (4) never occur.

If they do, the exceptional points have **extraordinarily** large heights (e.g. $10^{10^{200}}$ for $X_{\text{ns}}^+(11^2)(\mathbb{Q})$).

label	level	notes	j -invariants/models of exceptional points
16.64.2.1	2 ⁴	$N_{\text{ns}}(16)$	$-2^{18} \cdot 3 \cdot 5^3 \cdot 13^3 \cdot 41^3 \cdot 107^3 / 17^{16}$ $-2^{21} \cdot 3^3 \cdot 5^3 \cdot 7 \cdot 13^3 \cdot 23^3 \cdot 41^3 \cdot 179^3 \cdot 409^3 / 79^{16}$
16.96.3.335	2 ⁴	$H(4) \subsetneq N_{\text{sp}}(4)$	$257^3 / 2^8$
16.96.3.343	2 ⁴	$H(4) \subsetneq N_{\text{sp}}(4)$	$17^3 \cdot 241^3 / 2^4$
16.96.3.346	2 ⁴	$H(4) \subsetneq N_{\text{sp}}(4)$	$2^4 \cdot 17^3$
16.96.3.338	2 ⁴	$H(4) \subsetneq N_{\text{sp}}(4)$	2^{11}
32.96.3.230	2 ⁵	$H(4) \subsetneq N_{\text{sp}}(4)$	$-3^3 \cdot 5^3 \cdot 47^3 \cdot 1217^3 / (2^8 \cdot 31^8)$
32.96.3.82	2 ⁵	$H(8) \subsetneq N_{\text{sp}}(8)$	$3^3 \cdot 5^6 \cdot 13^3 \cdot 23^3 \cdot 41^3 / (2^{16} \cdot 31^4)$
25.50.2.1	5 ²	$H(5) = N_{\text{ns}}(5)$	$2^4 \cdot 3^2 \cdot 5^7 \cdot 23^3$
25.75.2.1	5 ²	$H(5) = N_{\text{sp}}(5)$	$2^{12} \cdot 3^3 \cdot 5^7 \cdot 29^3 / 7^5$
7.56.1.2	7	$\subsetneq N_{\text{ns}}(7)$	$3^3 \cdot 5 \cdot 7^5 / 2^7$
7.112.1.2	7	$-I \notin H$	$y^2 + xy + y = x^3 - x^2 - 2680x - 50053$ $y^2 + xy + y = x^3 - x^2 - 131305x + 17430697$
11.60.1.3	11	$\subsetneq B(11)$	$-11 \cdot 131^3$
11.120.1.8	11	$-I \notin H$	$y^2 + xy + y = x^3 + x^2 - 30x - 76$
11.120.1.9	11	$-I \notin H$	$y^2 + xy = x^3 + x^2 - 2x - 7$
11.60.1.4	11	$\subsetneq B(11)$	-11^2
11.120.1.3	11	$-I \notin H$	$y^2 + xy = x^3 + x^2 - 3632x + 82757$
11.120.1.4	11	$-I \notin H$	$y^2 + xy + y = x^3 + x^2 - 305x + 7888$
13.91.3.2	13	$S_4(13)$	$2^4 \cdot 5 \cdot 13^4 \cdot 17^3 / 3^{13}, -2^{12} \cdot 5^3 \cdot 11 \cdot 13^4 / 3^{13}$ $2^{18} \cdot 3^3 \cdot 13^4 \cdot 127^3 \cdot 139^3 \cdot 157^3 \cdot 283^3 \cdot 929 / (5^{13} \cdot 61^{13})$
17.72.1.2	17	$\subsetneq B(17)$	$-17 \cdot 373^3 / 2^{17}$
17.72.1.4	17	$\subsetneq B(17)$	$-17^2 \cdot 101^3 / 2$
37.114.4.1	37	$\subsetneq B(37)$	$-7 \cdot 11^3$
37.114.4.2	37	$\subsetneq B(37)$	$-7 \cdot 137^3 \cdot 2083^3$

Table 1. All known exceptional groups, j -invariants, and points of prime power level.

Approach

Compute the set of **arithmetically maximal** subgroups

H is arithmetically maximal if

- $X_H(\mathbb{Q})$ is finite and
- for every G such that $H \leq G$, $X_G(\mathbb{Q})$ is infinite.

For such H **compute equations** for X_H and $j_H: X_H \rightarrow X(1)$

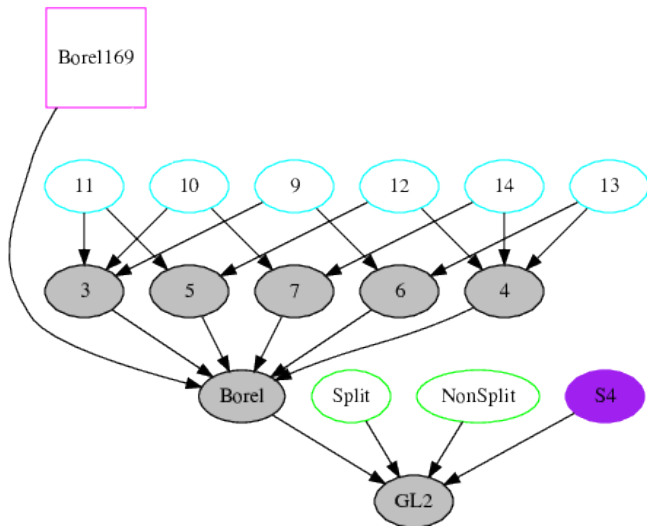
See Rouse's **VaNTAGe talk** for details and examples,

Assaf's **recent paper** and

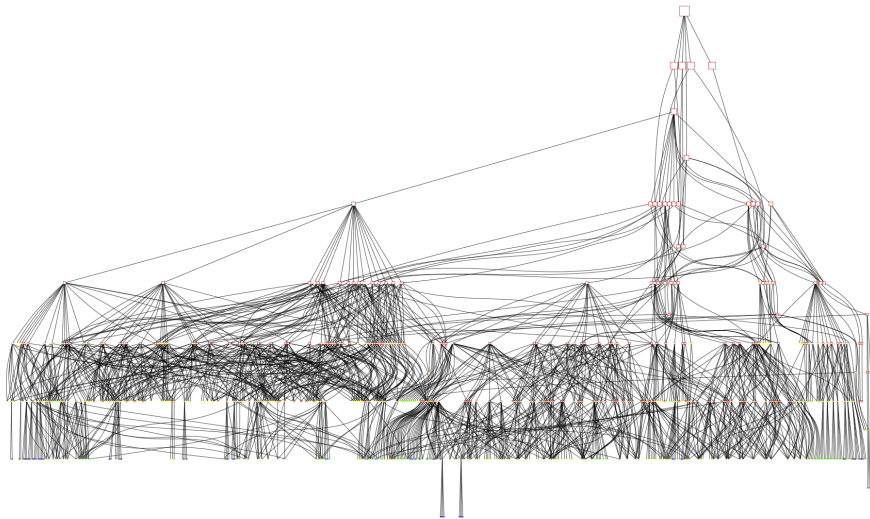
Zywina's **BIRS talk** for newer efficient approaches.

For such H with $-I \in H$ **determine the rational points** in $X_H(\mathbb{Q})$

Subgroups of $GL_2(\mathbb{Z}_{13})$



Subgroups of $GL_2(\mathbb{Z}_2)$



Explicit methods: highlight reel

- Local methods
 - Chabauty and Elliptic Chabauty
 - Mordell–Weil sieve
 - étale descent
 - Pryms
 - *Equationless étale descent via group theory*
 - *New techniques for computing $\text{Aut } C$*
-
- *Nonabelian Chabauty*
 - **“Equationless” local methods** and **Mordell–Weil sieve**
 - **Greenberg Transforms** (and big computations)
 - **Novel variants of existing techniques**
 - **Modularity of isogeny factors of J_H** (w/ Voight)

U N S O L V E D
mysteries

Arithmetically maximal level ℓ^n groups with $\ell \leq 13$ with $X_H(\mathbb{Q})$ **unknown**.

label	level	group	genus
27.243.12.1	3^3	$N_{\text{ns}}(3^3)$	12
25.250.14.1	5^2	$N_{\text{ns}}(5^2)$	14
49.1029.69.1	7^2	$N_{\text{ns}}(7^2)$	69
49.147.9.1	7^2	$G_{\text{ns}}^{\#}(7^2)$	9
49.196.9.1	7^2	$G_{\text{sp}}^{\#}(7^2)$	9
121.6655.511.1	11^2	$N_{\text{ns}}(11^2)$	511

Each has **rank = genus**, **rational CM points**, **no rational cusps**, and **no known exceptional points**.



Arithmetically maximal level ℓ^n groups with $\ell \leq 13$ with $X_H(\mathbb{Q})$ **unknown**.

label	level	group	genus
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Each has **rank = genus**, **rational CM points**, **no rational cusps**, and **no known exceptional points**.

²Balakrishnan–Betts–Hast–Jha–Müller

³Furio–Lombardo

Stacks and ramification

Some related facts:

- The map $X_{\text{ns}}(\ell^n) \rightarrow X(1)$ is ramified with degree ℓ^n at each cusp.
- Thus $X_{\text{ns}}(\ell^n) \rightarrow X_{\text{ns}}(\ell^{n-i})$ is ramified with degree ℓ^i at each cusp.
- The denominator of $j(E)$ is a perfect ℓ^n th power (for E a point on $X_{\text{ns}}(\ell^n)$).
- The denominator of the j -map $X_{\text{ns}}(\ell^n) \rightarrow X(1)$ is a perfect ℓ^n th power.

Resultants

The numerator and denominator of the j -map might not be coprime.
Furio–Lombardo: compare denominators of $X_{\text{ns}}(\ell)X(1)$ and $X_{\text{ns}}(\ell^2) \rightarrow X(1)$.

Denominator of j -map

Resultants

The numerator and denominator of the j -map might not be coprime.
Furio–Lombardo: compare denominators of $X_{\text{ns}}(\ell)X(1)$ and $X_{\text{ns}}(\ell^2) \rightarrow X(1)$.

Furio–Lombardo: Let $E/\mathbb{Q}$ be an elliptic curve without CM.

If $\text{Im } \rho_{E,49} \subseteq C_{\text{ns}}^+(49)$, there exist $k \in \{1, 8\}$ and pairwise coprime integers $x, y, z \in \mathbb{Z}$ such that

$$x^3 - 7x^2y + 7xy^2 + 7y^3 = k \cdot z^7 \quad \text{and} \quad j(E) = j_{\text{ns}} \left(\frac{x}{y} \right).$$

If $\text{Im } \rho_{E,49} \subseteq G_{\text{ns}}^{\#}(49)$, there exist $k \in \{1, 8\}$ and pairwise coprime integers $x, y, z \in \mathbb{Z}$ such that

$$x^3 - 7x^2y + 7xy^2 + 7y^3 = 7k \cdot z^7 \quad \text{and} \quad j(E) = j_{\text{ns}} \left(\frac{x}{y} \right).$$

If $\text{Im } \rho_{E,49} \subseteq G_{\text{sp}}^{\#}(49)$, there exist $k \in \{1, 7\}$ and pairwise coprime integers $x, y, z, w \in \mathbb{Z}$ such that $y = w^7$ and

$$x^3 - 4x^2y + 3xy^2 + y^3 = k \cdot z^7 \quad \text{and} \quad j(E) = j_{\text{sp}} \left(\frac{x}{y} \right).$$

Darmon–Granville, Bennett–Dahmen and invariant theory

Darmon and Granville prove that such equations have finitely many primitive solutions. (Because of stacks)

Bennett and Dahmen turn some of these into Fermat equations.

Theorem (Furio–Lombardo)

The $(\star)$ -solutions of the equation

$$a^2 + 28b^3 = 27c^7$$

are precisely

$$(\pm 1, -1, -1), \quad (\pm 27, -3, -1), \quad (\pm 2521, -61, -1).$$

The Poonen–Schaefer–Stoll playbook

- $(a, b, c) \mapsto E_{a,b,c}$
- $\exists E$ with $N_E \nmid N$ such that $E_{a,b,c}[7] \cong E[7]$
- This corresponds to a point $(E_{a,b,c}[7], \iota) \in X_E(7)$.
- $X_E(7)$ is a twist of the Klein quartic, a genus 3 plane quartic.
- One do Chabauty and more with this finite list of curves.

Theorem (ZB–Santiago Arango-Piñeros, **in progress**)

$X_{\text{ns}}^{\#}(49)$ and $X_{\text{sp}}^{\#}(49)$ *have no exceptional points*

Kummer: Fermat's Last Theorem for Regular primes

Theorem

If $(x, z, w) \in \mathbb{Z}^3$ is a solution to $x^7 + z^7 = w^7$, then 7 divides xzw .

Idea: factor over $L = \mathbb{Q}(\zeta_7)$, and use these facts about L and $K = \mathbb{Q}(\zeta_7 + \zeta_7^{-1}) \subset \mathbb{R}$:

- 1 $\mathbb{Q}(\zeta_7)$ has class number 1
- 2 $\mathcal{O}_L^\times = \mu_L \mathcal{O}_K^\times$
- 3 for all $\beta \in L$, $\beta^7 = b \pmod{7\mathcal{O}_L}$ for some $b \in \mathbb{Z}$.

Let $L = \mathbb{Q}(\zeta_7)$ and $K = \mathbb{Q}(\zeta_7 + \zeta_7^{-1}) \subset \mathbb{R}$. Then

$$x^7 + z^7 = \prod (x - \zeta_7^i z) = w^7$$

Since $\text{cl}(L) = 1$ and $\mathcal{O}_L^\times = \mu_L \mathcal{O}_K^\times$

$$x - \zeta_7 z = (\zeta_7^r u) \beta^7, \quad \text{for } u \in \mathcal{O}_K^\times, \quad \beta \in \mathcal{O}_L.$$

Elementary: $\beta^7 \equiv b \pmod{7\mathcal{O}_L}$ for some $b \in \mathbb{Z}$. Thus

$$x - \zeta_7 z = \zeta_7^r u b$$

Applying complex conjugation gives

$$x - \zeta_7^{-1} z = \zeta_7^{-r} u b$$

Combining

$$\zeta_7^r (x - \zeta_7^{-1} z) \equiv \zeta_7^{-r} (x - \zeta_7 z) \pmod{7\mathcal{O}_L}$$

i.e.,

$$x \cdot 1 - z \cdot \zeta_7 + z \cdot \zeta_7^{2r-1} - x \cdot \zeta_7^{2r} \equiv 0 \pmod{7\mathcal{O}_L}.$$

If the elements $1, \zeta_7, \zeta_7^{2r-1}, \zeta_7^{2r}$ are all distinct, then they are part of a $\mathbb{Z}$ basis for $\mathcal{O}_L = \mathbb{Z}[\zeta_7]$, and so implies that $7 \mid x$ and $7 \mid z$.

A “variant” of Kummer’s proof

Suppose that we want to solve

$$x^3 - 4x^2z + 3xz^2 + z^3 = w^7.$$

Kummer’s approach gives $u \in \mathcal{O}_L^*/(\mathcal{O}_L^*)^7$ such that

$$x - z\alpha = u\beta^7$$

Extend $\{1, \alpha\}$ to a basis. (α^2 will do fine.) Write

$$\beta = a + b\alpha + c\alpha^2$$

where a, b, c are variables. Then

$$x - z\alpha = \beta^7 = (a + b\alpha + c\alpha^2)^7 = f_0(a, b, c) + f_1(a, b, c)\alpha + f_2(a, b, c)\alpha^2$$

for some homogenous degree 7 polynomials f_i . In particular,

$$x = f_0(a, b, c)$$

$$-z = f_1(a, b, c)$$

$$0 = f_2(a, b, c).$$

This last equation $f_2(a, b, c)$ defines a smooth genus 15 curve C .

$$x - z\alpha = \beta^7 = (a + b\alpha + c\alpha^2)^7 = f_0(a, b, c) + f_1(a, b, c)\alpha + f_2(a, b, c)\alpha^2$$

$f_2(a, b, c)$ defines a smooth genus 15 curve C .

Over $K = \mathbb{Q}(\zeta_7)$ we have computable maps

$$C \rightarrow D \rightarrow E$$

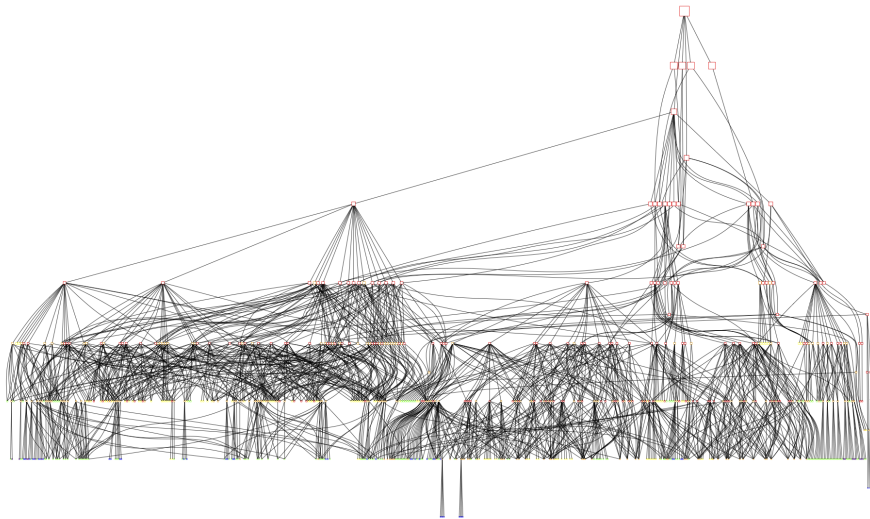
where D is a genus 3 curve and E is an elliptic curve with $\text{rk } E(K) = 0$.

For $u = 1$ this took most of a day, and a longer computation is running now to verify all of the cases.

If we pick a better basis, this works **integrally** and “recovers” Kummer’s proof, and generally gives lots of “divisibility” information.

E.g. for $u = 1$, $f_1(a, b, c)$ is a multiple of 7, and so z is too.

Subgroups of $GL_2(\mathbb{Z}_2)$



Agreeable groups

Zywina (indices occurring infinitely often, modulo conjectures)

The **index** of $\rho_{E,N}(G_{\mathbb{Q}})$ divides 220, 336, 360, 504, 864, 1152, 1200, 1296 or 1536.

Remark

$j(E) \in \{-7 \cdot 11^3, -7 \cdot 137^3 \cdot 2083^3\}$ *does occur finitely often*

Definition

An open $G \leq \mathrm{GL}_2(\widehat{\mathbb{Z}})$ is **agreeable** if $\det G = \widehat{\mathbb{Z}}^\times$, contains all scalar matrices, and the levels of G and $G \cap \mathrm{SL}_2(\widehat{\mathbb{Z}})$ are divisible by the same odd prime divisors.

Zywina computes a finite list of agreeable G such that determining $X_G(\mathbb{Q})$ determines all possibilities for the index.

Agreeable groups

Zywina computes a finite list of agreeable G such that determining $X_G(\mathbb{Q})$ determines all possibilities for the index.

Such G are **arithmetically maximal**, i.e., $X_G(\mathbb{Q})$ is finite and for every H such that $G \leq H$, $X_H(\mathbb{Q})$ is infinite.

If G is normal in H then the map of coarse spaces

$$X_G \rightarrow X_H$$

factors through a root stack $\mathcal{X}$ over X_H .

$\mathcal{X}$ is a stacky $\mathbb{P}^1$ or stacky elliptic curve, and is hyperbolic.

One can thus work with (twists of) étale covers of $\mathcal{X}$ (e.g., Belyi covers) and mostly ignore G .

If G is not normal in H , one can look directly at the ramification and might get lucky.

Root stacks

Let $D \subseteq X$ be an effective divisor on a scheme and $r \in \mathbb{Z}_{\geq 1}$.

The r th root stack $\sqrt[r]{(X, D)}$ of (X, D) :

$$\begin{array}{ccc} & \sqrt[r]{(X, D)} & \\ g \nearrow & \downarrow & \\ T & \xrightarrow{f} & X \end{array}$$

lifting f to g is the same as giving a divisor E on T and a linear equivalence $rE \sim f^*D$.